

# Alvin Reji

Boston, MA | (516) 424 0863 | [reji.alv@northeastern.edu](mailto:reji.alv@northeastern.edu) | [linkedin.com/in/alvin-reji/](https://www.linkedin.com/in/alvin-reji/) | [alvinreji.com](https://alvinreji.com)

## EDUCATION

**Northeastern University**, Boston, MA

Sep 2024 – Dec 2027

*Candidate for Bachelor of Science in Mechanical Engineering with an Aerospace Minor*

GPA: 3.84

**Activities:** AerospaceNU, ASME, Materials Engineering Research, TAMID, InterVarsity

**Relevant Courses:** Statics, Mechanics of Materials, Dynamics, Propulsion, Mechanical Design, Thermodynamics, Fluid Mechanics, Heat Transfer, Computational Fluid Dynamics, Systems Analysis & Control, Electrical Engineering, Mat Science, Engineering Computations

## SKILLS

**Hardware:** Microcontrollers, CNC Routing, 3D Printing, TIG Welding, Circuit Design, Soldering, Laser Cutting, Oscilloscope, ShopBot

**Software:** SolidWorks, Ansys, Minitab, MS Office, GSuite, Altium, MATLAB, AutoCAD, Python, C++, Creo, HTML, MagicDraw, PLM

**Interests:** Rock climbing, Weightlifting, Volleyball, Spikeball, Cycling, Running, Hiking, Billiards, Photography, Videography, Pottery

## EXPERIENCE

**Massachusetts Institute of Technology**

**Cambridge, MA**

*Teaching Assistant*

Aug 2025 – Present

- Educated 54 students on engineering design, mechanics, manufacturability, and iterative prototyping through design projects
- Fostered a collaborative, inclusive classroom by coaching conflict resolution, and promoting effective decision-making strategies
- Led student teams through design projects by providing structured guidance on engineering problem-solving and teamwork

**BAE Systems**

**Nashua, NH**

*Mechanical Engineering Intern*

June 2026 – Aug 2026

- Developed Block Definition and Use Case Diagrams in MagicDraw to define system integration and user-product interactions
- Designed locking mechanisms and cable management solutions for complex systems within classified laboratory environments
- Executed root-cause analyses by developing Linux-based data logging tools and utilizing statistical analysis to determine factors
- Led hardware development for multi-point RF geo-locator system, whilst collaborating cross-functionally with engineering teams
- Designed and fabricated IP53-rated, dynamic antenna modules using Creo and FDM printing to identify and locate ally signals

**Johnson & Johnson Medtech (Abiomed)**

**Danvers, MA**

*Product Development Engineering Co-Op*

Jan 2026 – June 2026

- Designed and iterated custom SolidWorks components for Instron testing using FDM/SLA printing for product verification tests
- Conducted 100+ mechanical tests simulating clinical-use conditions and optimized test method to improve variability by 73%
- Performed statistical analyses using Minitab to execute TMVs and Gage R&R studies for data-driven design improvements
- Developed a Python/OpenCV pipeline to automate analyses across 30+ experimental iterations, improving design efficiency
- Authored engineering documentation, test protocols, and work instructions ensuring compliancy, traceability, and consistency

**CubeSAT at Northeastern University**

**Boston, MA**

*Mechanical Engineer*

Sep 2024 – Dec 2025

- Designed 1U CubeSat optical component housings using FEA and DFMA principles to withstand mission-specific structural loads
- Generated ASME Y14.5-compliant engineering drawings applying GD&T to control critical features for fabrication and assembly

**AerospaceNU Rocket Team at Northeastern University**

**Boston, MA**

*Avionics Engineer*

Sep 2024 – Dec 2025

- Developed software for reliable sensor data acquisition on flight computers by implementing signal processing and noise reduction
- Designed flight computer PCBs optimized for power distribution, EMI compliance, and thermal management using KiCAD

**University Nanosatellite Program at University at Buffalo**

**Buffalo, NY**

*Guidance, Navigation, & Control Engineer*

Aug 2023 – May 2024

- Developed MATLAB simulations for satellite missions, modeling orbital mechanics, attitude dynamics, and control responses
- Implemented Monte Carlo algorithms and Kalman filters to improve satellite state determination and reduce signal noise by 54%
- Presented mission architecture, GNC simulation results, and subsystem performance analyses during CDR/PDR to AFRL experts

## PROJECTS

**GE1501 Cornerstone of Engineering at Northeastern University**

**Boston, MA**

*Horse Racing Tracking Collar*

Oct 2024 – Dec 2024

- Designed and fabricated an ESP32-based collar using SolidWorks and FDM 3D printing, ensuring sensor stability and durability
- Implemented GPS and accelerometer integration, enabling accurate horse position and motion tracking with  $\pm 1$  m positional error
- Developed embedded C++ software for real-time Wi-Fi data transmission and data visualization through a custom PyQt5 GUI

**NASA Techrise Student Challenge**

**Elmont, NY**

*Atmospheric Measurement of Noxious Gases Using Sensors (AMONGUS)*

Nov 2021 – July 2022

- Conducted pre-launch testing in low-pressure chambers, improving payload reliability by 30% under high altitude conditions
- Fabricated and assembled payload components using CNC machining and laser cutting, ensuring structural integrity during flight
- Performed system integration, calibration, and validation of environmental sensors, ensuring accurate flight data collection